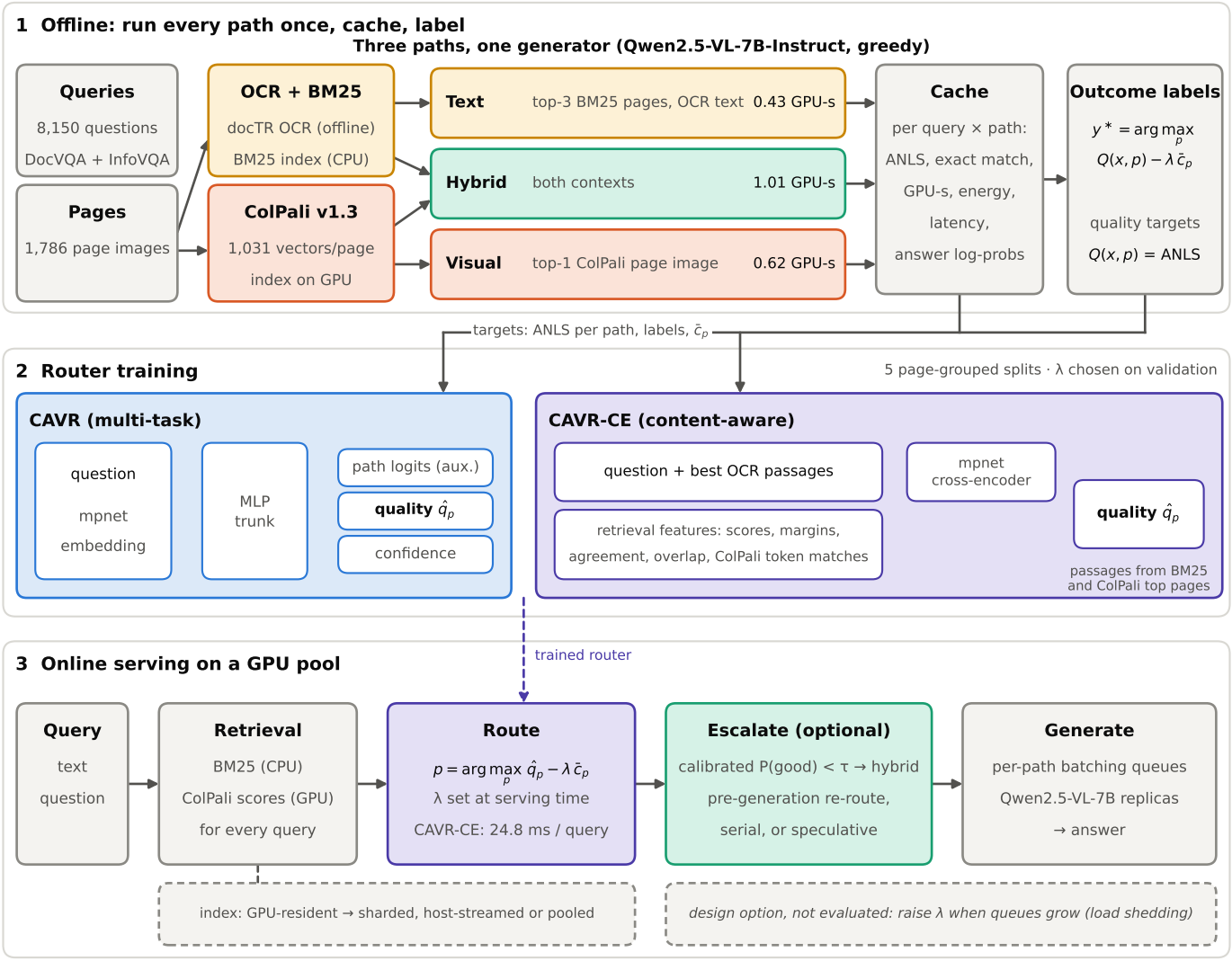

1 Offline: run every path once, cache, label
Three paths, one generator (Qwen2.5-VL-7B-Instruct, greedy)
Queries
8,150 questions
DocVQA + InfoVQA
Pages
1,786 page images
OCR + BM25
docTR OCR (offline)
BM25 index (CPU)
ColPali v1.3
1,031 vectors/page
index on GPU
Text
top-3 BM25 pages, OCR text
0.43 GPU-s
Hybrid
both contexts
1.01 GPU-s
Visual
top-1 ColPali page image
0.62 GPU-s
Cache
per query × path:
ANLS, exact match,
GPU-s, energy,
latency,
answer log-probs
Outcome labels
$y^* = \arg\max_p Q(x,p) - \lambda \tilde{c}_p$
quality targets
$Q(x,p)$ = ANLS
targets: ANLS per path, labels, $\tilde{c}_p$
2 Router training
5 page-grouped splits · λ chosen on validation
CAVR (multi-task)
question
mpnet
embedding
MLP
trunk
path logits (aux.)
quality $\hat{q}_p$
confidence
CAVR-CE (content-aware)
question + best OCR passages
mpnet
cross-encoder
retrieval features: scores, margins,
agreement, overlap, ColPali token matches
quality $\hat{q}_p$
passages from BM25
and ColPali top pages
trained router
3 Online serving on a GPU pool
Query
text
question
Retrieval
BM25 (CPU)
ColPali scores (GPU)
for every query
Route
$p = \arg\max_p \hat{q}_p - \lambda \tilde{c}_p$
λ set at serving time
CAVR-CE: 24.8 ms / query
Escalate (optional)
calibrated P(good) < τ → hybrid
pre-generation re-route,
serial, or speculative
Generate
per-path batching queues
Qwen2.5-VL-7B replicas
→ answer
index: GPU-resident → sharded, host-streamed or pooled
design option, not evaluated: raise λ when queues grow (load shedding)